ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY
UNIVERSITY FOR INDUSTRY

MEng2101

Engineering Drawing

College of Engineering

Department of Mechanical engineering

Course Outline

1st Semester 2026/ 27

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1. Course Information

Course Code	MEng2101	
Course Name	Engineering Drawing	
Year/Semester	Year 2/I	
Academic Year	2026/2027	
Pre-requisite(s)	Nil	
Co-requisite(s)	Nil	
Credit Hour/ ECTS	3/5 (1 Lecture and 6 Practical hours per week)	
Course schedule	Mechanical Sec1	Lec: Wednesday 8:30-11:20: B-48/ DR-2 Prac: Friday 1:30- 4:20: B-47/ DR-1
	Mechanical Sec2	Lec: Friday 8:30-11:20: B-47/ DR-1 Pract: Thursday 7:30-4:20: B-51/DR-1

2. Instructor Information

Course Coordinator

Name	[xxx xxx]
E-mail	[xxx.xxx]@aastu.edu.et
Telephone number	+251[xxxxxxxxx]
Consultation hour	[day xx:xx -xx:xx]
Office (Block/Room)	[xx/xxx]

Course Instructor

Name	Tolina Tabo
E-mail	tolina.tabo@aastu.edu.et
Telephone number	+251949462523
Consultation hour	Wednesday 10:30 a.m- 12:30 p.m Thursday 2:00 4:00 p.m
Office (Building/Room)	B-64/ Room 304

3. Course Overview

The Engineering Drawing course is designed to introduce students to the fundamental principles and practices of technical drawing, which is a crucial skill in the field of engineering. This course covers a wide range of topics, from the basic concepts and importance of engineering drawings to the use of various drawing instruments, standards, and conventions. Students will learn both theoretical and practical aspects of engineering drawing, including different types of projections, multi-view drawings, pictorial drawings, auxiliary views, and sectional views.

4. Course Objectives

The objective of the Engineering Drawing course is to equip students with the fundamental skills and knowledge necessary to create, interpret, and utilize engineering drawings effectively. This course aims to foster a deep understanding of technical drawing principles, enhance visualization abilities, and develop proficiency with traditional drawing instruments. The course also prepares students to communicate technical information clearly and accurately, which is essential for success in various engineering disciplines.

5. Course Learning Outcomes

Up on successful completion of this course, students should be able to:		Program learning outcomes
CLO1	Interpret and visualize engineering drawings, drawing conventions and standards	PLO1
CLO 2	Draw projection of point, line, plane and solid	PLO3
CLO 3	Construct multi-view drawings or supplement views from real objects or given pictorial drawings using appropriate engineering drawing tool and techniques.	PLO3
CLO 4	Draw pictorial projections and drawing from real object or given multiview drawings considering standard features	PLO3
CLO 5	Function effectively as an individual and as group member in assignments and course activities.	PLO9

6. Graduate Attributes

PLO1	Engineering Knowledge		PLO7	Environment and sustainability
PLO2	Problem Analysis		PLO8	Ethics
PLO3	Design/development of solutions		PLO9	Individual and teamwork
PLO4	Investigation		PLO10	Communication
PLO5	Modern tool usage		PLO11	Project management and finance
PLO6	The engineer and society		PLO12	Lifelong learning

7. Course Schedule and Instructional Method

This course consists of 98 hours of class contact hours. You are expected to take an additional 22 hours of non class contact hours to complete assessments, readings and exam preparation.

In the Learning and Teaching Activity column the learning and teaching activities used in the course includes lectures supported with tutorials and practical exercises. The students to participate and learn in these various components in face-to-face and other learning mode. These learning activities engage students in using various resources and working collaboratively on tasks, ultimately leading them to achieve the course learning outcomes.

By participating in hands-on experiences, collaborating with peers, and reflecting on their learning, students will develop the skills and knowledge necessary for success in engineering drawing and the broader engineering discipline.

Week	Topic [Module]	Learning and Teaching Activity	CLO
Week 1	**1. Introduction to Engineering Drawing** • History of Engineering Drawing • Objective of Engineering Drawing • Lettering • Drawing Instrument	- Guided learning: Lec -1 hrs & Pract -6 hrs, - Guided learning: NF2F-1hr - Independent- NF2F-1hrs	CLO1
Week 2	**1. Introduction to Engineering Drawing** • Types of Lines • Basic Geometric Construction	- Guided learning: Lec -1 hr & Pract -6 hrs, - Guided learning: NF2F-1 hr - Independent- NF2F-1hr	CLO1
Week 3	**2. Theory of Projection** • Classifications of Projections • Projection of Point	- Guided learning: Lec -1 hr & Pract -6 hrs, - Guided learning: NF2F-1 hr - Independent- NF2F-1hr	CLO1 CLO2
Week 4	**2. Theory of Projection** • Projection of Line • Projection of Plane	- Guided learning: Lec -1 hr & Pract -6 hrs, - Guided learning: NF2F-1 hr - Independent- NF2F-1 hr	CLO1 CLO2
Week 5	**3. Multi-View Drawings** • System of Projection • Choice of Views • One view Drawings	- Guided learning: Lec -1 hr & Pract -6 hrs, - Guided learning: NF2F-0.5 hr - Independent- NF2F-0.5 hr	CLO1 CLO3
Week 6	**3. Multi-View Drawings** • Two view, and three view Drawings • Laying Out of Drawings	- Guided learning: Lec -1 hr & Pract -6 hrs, - Guided learning: NF2F-0.5 hr Independent- NF2F-0.5 hr	CLO1 CLO3 CLO5
Week 7	**3. Multi-View Drawings** • Width, Height, and Depth relation ships • Precedence of Lines	• Guided learning: Lec -1 hr & Pract -6 hrs, • Guided learning: NF2F-0.5 hr Independent- NF2F-0.5 hr	CLO1 CLO3 CLO5
Week 8	**3. Multi-View Drawings** • Practical Exercises	- Guided learning: Pract -7 hrs, - Guided learning: NF2F-0.5 hr - Independent- NF2F-0.5 hr	CLO1 CLO3 CLO5

Week 9	**4. Pictorial Drawing** • Comparison between Multi view and • Pictorial drawing • Types of Pictorial Projections	- Guided learning: Lec -1 hrs & Pract -6 hrs, - Guided learning: NF2F-0.5hr - Independent- NF2F-0.5hr	CLO1 CLO4
Week 10	**4. Pictorial Drawing** • Isometric Drawings	- Guided learning: Lec -1 hrs & Pract -6 hrs, - Guided learning: NF2F-0.5hr - Independent- NF2F-0.5hr	CLO1 CLO4 CLO5
Week 11	**4. Pictorial Drawing** • Oblique Drawings	- Guided learning: Lec -1 hrs & Pract -6 hrs, - Guided learning: NF2F-0.5hr - Independent- NF2F-0.5hr	CLO1 CLO4 CLO5
Week 12	**4. Pictorial Drawing** • Practical Exercises	- Guided learning: Pract -7 hrs, - Guided learning: NF2F-0.5hr - Independent- NF2F-0.5hr	CLO1 CLO4 CLO5
Week 13	**5. Auxiliary Views** • Primary Auxiliary Views • Secondary Auxiliary Views • Complete and partial Auxiliary Views	- Guided learning: Lec -1 hrs & Pract-6 hrs, - Guided learning: NF2F-1hr - Independent- NF2F-1hr	CLO1 CLO3
Week 14	**6. Sectional View** • Types of Sections • Making Sectional views • Conventional Representations • Sectional Auxiliary Views	- Guided learning: Lec -1 hr & Pract-6 hrs, - Guided learning: NF2F-1 hr - Independent- NF2F-1 hr	CLO1 CLO3

8. Assessment Tasks

No	Assessment Method	Weight	Week Due	CLO Assessed
1	Quizzes	5%	Weeks 4, 7, 10, and 14	CLO1, 2, 3, and 4
2	Test	25%	Week 8 and 14	CLO1, 2, 3, and 4
3	Assignment	20%	Week 2 to 14	CLO1, 2, 3, 4 and 5
4	Final Exam	50%	Week 15 or 16	CLO1, 2, 3, 4 and 5

Submission of Assessment Tasks

Late classworks/ assignments will result in a grade reduction of 10% per 24 hours late (or portion thereof).

Feedback on Assessments

Students receive feedback on their assessment activities and/or marked submissions for each task. The feedback is given to students after the evaluation of each assessment in the classroom or office.

9. Learning Resources

Textbooks

1. Cecil Jensen and Jay D. Helsel, Engineering Drawing and Design, McGraw-Hill, New York, 7th Ed, 2008.

References

1. Thomas French, Charles Vierck, and Robert Foster, Engineering Drawing and Graphics Technology, 1993
2. Warren J. Luzzader and Jon M. Duff, Fundamentals of engineering drawing, Prentice Hall, New Jersy, 11thed, 1983.
3. R.S.Vaishwanar, Engineering Drawing and Graphics, Kumar Offset Press, New Delhi, 1993.
4. 5 Voland, Gerard G.S., Modern Engineering Drawing and Graphics and Designs, West Publishing Company, 1987.

Online Resources

Students will use online resources such as elearning, telegram, google.com and youtube.com.

10. Grading Policy

Grading of courses will be according to the university's rules and regulation. Examinations are graded and the letter grading system with corresponding points shown below.

Marks	Grade	Grade Points
[90,100]	**A^+**	4.00
[85,90)	**A**	4.00
[80,85)	**A^-**	3.75
[75,80)	**B^+**	3.50
[70,75)	**B**	3.00
[65,70)	**B^-**	2.75
[60,65)	**C^+**	2.50
[50,60)	**C**	2.00
[50	**F**	0.00

Pass requirements

To pass this course, a student must:

- Attempt all assessments.
- Achieve a minimum of 40% in the final exam.

11. Attendance

As per university guideline, a minimum 80% during lecture and 100% during practical work sessions except for some unprecedented mishaps. Failure to fulfil this requirement results in barring.

12. Academic Integrity, Referencing and Plagiarism

Academic Integrity

AASTU values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offenses under the Code of Student Conduct and Disciplinary Procedures. Work submitted must be your own, and others' ideas should be appropriately acknowledged. If you don't follow these rules, plagiarism may be detected in your work using plagiarism detection tool 'Turnitin'.

Referencing

Referencing is a way of acknowledging the sources of information that you use to research your assignments. You need to provide a reference whenever you draw on someone else's words, ideas or research. Not referencing other people's work can constitute plagiarism.

Referencing Style

The University advises students to use the "IEEE Referencing Style" for written work and oral presentations.